

Supplementary information

Tools for the shifted-zeta Wilson–Loewner program

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20 September 2026 – Version 0.3

Abstract

This companion to *Shifted-zeta evolution and Wilson lines* collects the detailed tools used to formulate that research program. It gives the causal arithmetic factorization, the free defect kernel and auxiliary gamma model, the constant-driver contour dictionary, the exact smooth Wilson variation, chiral defect projectors, endpoint selection rules, and ordinary reflected pairing. A smooth junction and its direct Gaussian bulk response provide a controlled deformation calculation. We also give the algebraic cumulative-storage gluing and simultaneous depth–shift criteria used in the exposition. The established contour and free-field calculations do not identify the arithmetic transfer or prove interacting reflection positivity. The three preserved diagnostic programs contain 268 finite cases.

How to read this companion. The program, its physical realization hypothesis and the route to RH are developed in the main exposition [1]. Sections S1–S8 retain the technical calculations of the preceding unified manuscript, with arithmetic conventions repeated for independent reading. Section S9 records the inherited continuation algebra; Section S10 describes the diagnostics. The two documents have separate numbering and bibliographies and can be compiled independently. References to numbered sections below are local unless explicitly attributed to the exposition.

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S1 Arithmetic conventions and exact kernel factorization

We repeat the arithmetic normalization to make this companion independently readable. The effective-evolution hypothesis and its role in the program are stated in the exposition [1]. The free defect calculations below use the exponential tower

$$n_{\Gamma}(u) = \frac{e^{-u/2}}{1 - e^{-2u}} = \sum_{n=0}^{\infty} e^{-(2n+1/2)u}, \quad u > 0. \quad (\text{S1.1})$$

S1.1 The compactly supported Weil form

Let $I_L = (-L/2, L/2)$, with $L > 0$, and let $f \in C_c^\infty(I_L; \mathbb{C})$. Write F for its zero extension to $\mathbb{R}$ and use

$$\widehat{F}(\tau) = \int_{\mathbb{R}} e^{-i\tau x} F(x) dx, \quad \langle f, g \rangle = \int_{I_L} \overline{f(x)} g(x) dx. \quad (\text{S1.2})$$

For $a > 0$, define the compressed shift $T_a f(x) = F(x - a)$ on I_L . Let $\psi = \Gamma'/\Gamma$ be the digamma function, and let $\Lambda(n) = \log p$ when $n = p^k$, p prime and $k \geq 1$, and zero otherwise. Finally put

$$C(f) = \int_{I_L} f(x) \cosh(x/2) dx, \quad S(f) = \int_{I_L} f(x) \sinh(x/2) dx. \quad (\text{S1.3})$$

The normalization of the program is

$$\begin{aligned} Q_{0,L}[f] &= \frac{1}{2\pi} \int_{\mathbb{R}} [\operatorname{Re} \psi(\tfrac{1}{4} + \tfrac{i\tau}{2}) - \log \pi] |\widehat{F}(\tau)|^2 d\tau + 2|C(f)|^2 - 2|S(f)|^2 \\ &\quad - \sum_{\substack{n \geq 2 \\ \log n < L}} \frac{\Lambda(n)}{\sqrt{n}} \langle f, (T_{\log n} + T_{\log n}^*) f \rangle. \end{aligned} \quad (\text{S1.4})$$

The prime-power sum is finite on a fixed interval. The pole term is indefinite, and the archimedean multiplier includes its local constant. Neither may be dropped in a positivity argument.

For comparison with the usual explicit formula, set $h = F * \bar{F}$, where $\bar{F}(x) = \overline{F(-x)}$. The two prime sums in the Weil functional give the last line of (S1.4); its $e^{x/2} + e^{-x/2}$ term gives $2|C|^2 - 2|S|^2$. Its archimedean distribution has Fourier multiplier $\operatorname{Re} \psi(1/4 + i\tau/2) - \log \pi$. Thus $Q_{0,L}[f] = W(F * \bar{F})$ in the convention of [11, p. 1]. In particular the weight is $\Lambda(n)/\sqrt{n}$.

Theorem S1.1 (Weil positivity criterion, used as background). *RH holds if and only if $Q_{0,L}[f] \geq 0$ for every $L > 0$ and every $f \in C_c^\infty(I_L; \mathbb{C})$.*

The equivalence is the classical criterion, not a result of the present construction. Every compact support lies in one such interval, so an unbounded sequence of interval lengths suffices. A finite collection of lengths or numerical matrices does not.

S1.2 The positive kinetic term and the indispensable local term

Put $a_n = 2n + 1/2$, for $n \geq 0$, and

$$b(\tau^2) = \operatorname{Re} \psi(\tfrac{1}{4} + \tfrac{i\tau}{2}) - \psi(\tfrac{1}{4}) = 2 \sum_{n=0}^{\infty} \frac{\tau^2}{a_n(a_n^2 + \tau^2)} \geq 0, \quad w_0 = \psi(\tfrac{1}{4}) - \log \pi < 0. \quad (\text{S1.5})$$

The constant $\psi(1/4)$ is not equal to $\log \pi$. Adding and subtracting $\psi(1/4)$ in the original multiplier gives the exact identity

$$\begin{aligned} \operatorname{Re} \psi(\tfrac{1}{4} + \tfrac{i\tau}{2}) - \log \pi &= [\operatorname{Re} \psi(\tfrac{1}{4} + \tfrac{i\tau}{2}) - \psi(\tfrac{1}{4})] + [\psi(\tfrac{1}{4}) - \log \pi] \\ &= b(\tau^2) + w_0. \end{aligned} \quad (\text{S1.6})$$

Thus w_0 is the remainder after separating the nonnegative multiplier $b(\tau^2)$; it is already present in (S1.4), not an additional term introduced into the form.

The series in (S1.5) follows by subtracting the convergent digamma partial fractions. Each exponential in (S1.1) satisfies $\int_0^\infty e^{-au} (1 - \cos \tau u) du = \tau^2/[a(a^2 + \tau^2)]$. Plancherel therefore gives

$$\mathcal{E}_\Gamma[F] := \int_0^\infty n_\Gamma(u) \|F(\cdot + u) - F\|_2^2 du = \frac{1}{2\pi} \int_{\mathbb{R}} b(\tau^2) |\widehat{F}(\tau)|^2 d\tau. \quad (\text{S1.7})$$

Using (S1.6) and $\frac{1}{2\pi} \int_{\mathbb{R}} |\widehat{F}(\tau)|^2 d\tau = \|f\|_2^2$, the archimedean contribution becomes

$$\frac{1}{2\pi} \int_{\mathbb{R}} [\operatorname{Re} \psi(\frac{1}{4} + \frac{i\tau}{2}) - \log \pi] |\widehat{F}(\tau)|^2 d\tau = \mathcal{E}_{\Gamma}[F] + w_0 \|f\|_2^2. \quad (\text{S1.8})$$

Substituting this identity into (S1.4) rewrites the full target as

$$\begin{aligned} Q_{0,L}[f] &= \mathcal{E}_{\Gamma}[F] + w_0 \|f\|_2^2 + 2|C(f)|^2 - 2|S(f)|^2 \\ &\quad - \sum_{\substack{n \geq 2 \\ \log n < L}} \frac{\Lambda(n)}{\sqrt{n}} \langle f, (T_{\log n} + T_{\log n}^*) f \rangle. \end{aligned} \quad (\text{S1.9})$$

Here $w_0 \|f\|_2^2 = w_0 \int_{I_L} |f(x)|^2 dx$ is the fixed negative local term. The positive difference energy is only one part of the desired form: its local term, pole contribution and prime-power translations must all be retained.

The natural form domain on a fixed interval is given by zero extensions with $\int \log(2 + |\tau|) |\widehat{F}(\tau)|^2 d\tau < \infty$. Indeed $b(\tau^2)$ grows logarithmically at infinity, whereas the pole and finite translation terms are bounded forms on $L^2(I_L)$. This explains the logarithmic form domain and semiboundedness of the localized form. For the logical proof target in Theorem S1.1, the smooth compactly supported class is enough.

S1.3 Shifted transfers and a sufficient contraction route

Define the entire completed zeta function and its centered version by

$$\xi(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad H(p) = \xi(\frac{1}{2} + p). \quad (\text{S1.10})$$

For $0 < \omega \leq 1/2$, the shifted transfer is

$$K_{\omega}(p) = \frac{H(p - \omega)}{H(p + \omega)}. \quad (\text{S1.11})$$

The functional equation and reality imply $|K_{\omega}(i\tau)| = 1$ where the ratio is defined. This boundary identity is unconditional; it is not a proof that an unweighted convolution operator is a contraction. In particular poles away from that boundary cannot be ignored.

The causal interpretation uses inverse Laplace transform on a line $\operatorname{Re} p = \sigma > 1$. Let k_{ω} be that causal distribution and $V_{\omega,L} = P_L(k_{\omega}*)P_L$, where P_L denotes restriction to I_L together with zero extension when used as an input. The explicit kernel below is locally integrable for $\omega > 0$, so these finite-window operators are bounded. At $\omega = 0$, $k_0 = \delta_0$ and $V_{0,L} = I$. Logarithmic differentiation gives

$$\partial_{\omega} K_{\omega}(p) = -a_{\omega}(p) K_{\omega}(p), \quad a_{\omega}(p) = \frac{H'}{H}(p - \omega) + \frac{H'}{H}(p + \omega). \quad (\text{S1.12})$$

At zero shift the causal generator has Laplace symbol

$$a_0(p) = \frac{2}{p + 1/2} + \frac{2}{p - 1/2} - \log \pi + \psi(\frac{1}{4} + \frac{p}{2}) - 2 \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} e^{-p \log n}. \quad (\text{S1.13})$$

The identity

$$\psi(\frac{1}{4} + \frac{p}{2}) - \psi(\frac{1}{4}) = 2 \int_0^{\infty} n_{\Gamma}(u) (1 - e^{-pu}) du \quad (\text{S1.14})$$

shows that the real quadratic form of this causal generator is exactly (S1.4). The two rational terms give the symmetric kernel $2 \cosh((x - y)/2)$; its form is $2|C|^2 - 2|S|^2$. On smooth test functions the first variation is therefore

$$\langle f, (I - V_{\omega,L}^* V_{\omega,L}) f \rangle = 2\omega Q_{0,L}[f] + o(\omega), \quad \omega \downarrow 0. \quad (\text{S1.15})$$

One obtains this expansion by differentiating the causal kernel at $\omega = 0$ using (S1.13); smooth zero extensions make the subtracted difference integral convergent in L^2 on a finite window. Thus an independent proof of $\|V_{\omega,L}\| \leq 1$ for every L and shifts tending to zero would imply RH. We need only this sufficient implication here; no global unweighted realization of K_ω is assumed.

S1.4 The exact positive part, and what it leaves unresolved

Write $a = 1/2 - \omega$, $b = 1/2 + \omega$, $s = 1/2 + p$, and $\mathcal{L}(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$, distinguished from the von Mangoldt function. Direct factorization yields

$$K_\omega(p) = \frac{(p+a)(p-b)}{(p+b)(p-a)} \widetilde{K}_\omega(p), \quad \widetilde{K}_\omega(p) = \frac{\mathcal{L}(s-\omega)}{\mathcal{L}(s+\omega)}. \quad (\text{S1.16})$$

The beta integral and Euler product, for $\text{Re } p > b$, give

$$g_\omega(t) = \frac{2\pi^\omega}{\Gamma(\omega)} e^{-at} (1 - e^{-2t})^{\omega-1} \mathbf{1}_{t>0}, \quad (\text{S1.17})$$

$$c_n(\omega) = n^{\omega-1/2} \prod_{\ell|n, \ell \text{ prime}} (1 - \ell^{-2\omega}), \quad c_1 = 1, \quad (\text{S1.18})$$

$$\widetilde{k}_\omega = g_\omega * \sum_{n \geq 1} c_n(\omega) \delta_{\log n} \geq 0. \quad (\text{S1.19})$$

For example, the substitution $r = e^{-2t}$ in the Laplace transform of (S1.17) gives $\pi^\omega \Gamma((p+a)/2) / \Gamma((p+b)/2)$. For the zeta quotient the coefficient at a prime power ℓ^k is $\ell^{k\omega} (1 - \ell^{-2\omega})$ before multiplication by $n^{-1/2}$, proving (S1.18) by multiplicativity. Only finitely many shifts contribute on a fixed window. These formulas define positive locally finite measures, not normalized probability laws; the endpoint $a = 0$ in particular precludes an unqualified Beta($a/2, \omega$) probability interpretation.

Absorb one rational factor into the positive part:

$$\begin{aligned} \widehat{k}_\omega &= \widetilde{k}_\omega + 2a(e^a \cdot \mathbf{1}_{>0}) * \widetilde{k}_\omega \geq 0, \\ k_\omega &= \widehat{k}_\omega - 2b(e^{-b} \cdot \mathbf{1}_{>0}) * \widehat{k}_\omega. \end{aligned} \quad (\text{S1.20})$$

Thus the full kernel is a positive part minus an exponential average. This exact reduction was central to the parent Loewner investigation [14]. It neither makes k_ω positive nor bounds the norm of its finite-window convolution. The required cancellation is still the arithmetic problem.

The fractional-dimension interpretation follows from the radial beta integral

$$\int_{\mathbb{R}^d} (1 + |u|^2)^{-(p+b)/2} d^d u = \pi^{d/2} \frac{\Gamma((p+b-d)/2)}{\Gamma((p+b)/2)}. \quad (\text{S1.21})$$

For integer d this is an ordinary integral in its convergence region; its radial continuation at $d = 2\omega$ reproduces the gamma factor. It is not a continuously variable lattice dimension. To see a concrete obstruction to one natural lattice continuation, put $\vartheta(q) = \sum_{k \in \mathbb{Z}} q^{k^2}$ and $m = 1 + d$. The coefficient of q^3 in $\vartheta(q)^m$ is

$$8 \binom{m}{3} = \frac{4}{3} m(m-1)(m-2) = \frac{4}{3} (d^3 - d) < 0 \quad (0 < d < 1). \quad (\text{S1.22})$$

It cannot be a positive point count. This excludes that interpolation, not every positive kernel or analytic continuation. Neither (S1.21) nor a bare Loewner map supplies the prime comb and rational subtraction in (S1.19)–(S1.20).

S2 Why defect matter reproduces the archimedean exponents

The geometric reason for using three-dimensional defect matter can be seen without supersymmetry. Normalize a free massless scalar covariance on the defect to $|X - Y|^{-1}$; coupling and color constants will be restored when needed. Write $X = e^x \Omega$, $Y = e^y \Omega'$ with $\Omega, \Omega' \in S^2$, and rescale the field by $e^{x/2}$. With $u = x - y$ and $v = \Omega \cdot \Omega'$, the covariance becomes

$$C(u, v) = \frac{1}{\sqrt{2(\cosh u - v)}} = \sum_{\ell=0}^{\infty} e^{-(\ell+1/2)|u|} P_{\ell}(v), \quad u \neq 0. \quad (\text{S2.1})$$

This is the Legendre generating function with parameter $e^{-|u|}$.

Fix an axis $e \in S^2$ and $0 \leq \rho < 1$. The Poisson density on S^2 and its even part are

$$P_{\rho}(v) = \frac{1 - \rho^2}{4\pi(1 - 2\rho v + \rho^2)^{3/2}}, \quad h_{\rho}(\Omega) = \frac{1}{2}\{P_{\rho}(e \cdot \Omega) + P_{\rho}(-e \cdot \Omega)\}. \quad (\text{S2.2})$$

Integrating (S2.1) against $h_{\rho}(\Omega)h_{\rho}(\Omega')$ projects onto even angular degrees. Orthogonality of the Legendre polynomials gives

$$C_{\rho}(u) = \sum_{n=0}^{\infty} \rho^{4n} e^{-a_n|u|} = \frac{e^{-|u|/2}}{1 - \rho^4 e^{-2|u|}}, \quad a_n = 2n + \frac{1}{2}. \quad (\text{S2.3})$$

For $\rho < 1$ this is a bounded continuous positive-definite covariance. For $u \neq 0$ it tends to $n_{\Gamma}(|u|)$ as $\rho \uparrow 1$. The limit is singular at $u = 0$, so positivity of the regulated covariance cannot simply be passed to a finite contact-subtracted covariance.

Proposition S2.1 (The sign of the subtraction). *Let T_{ρ} be convolution by C_{ρ} on $\mathbb{R}$, and put $M_{\rho} = \widehat{C}_{\rho}(0)$. Then $M_{\rho}I - T_{\rho}$ is positive, and its quadratic form tends to $\mathcal{E}_{\Gamma}$ on smooth compactly supported functions. The opposite subtraction $T_{\rho} - M_{\rho}I$ tends to the negative of that form. Adding any finite constant multiple of the identity to the latter limit does not give a positive form on all smooth compactly supported functions.*

Proof. Fourier transformation gives

$$\begin{aligned} \widehat{C}_{\rho}(\tau) &= \sum_{n \geq 0} \rho^{4n} \frac{2a_n}{a_n^2 + \tau^2}, \\ M_{\rho} - \widehat{C}_{\rho}(\tau) &= 2 \sum_{n \geq 0} \rho^{4n} \frac{\tau^2}{a_n(a_n^2 + \tau^2)} \uparrow b(\tau^2). \end{aligned} \quad (\text{S2.4})$$

Monotone convergence and (S1.7) prove the first claims. Moreover $M_{\rho} = \sum_{n \geq 0} \rho^{4n} / (n + 1/4)$ diverges logarithmically. Since $b(\tau^2) \rightarrow \infty$, any multiplier $c - b(\tau^2)$ is negative at high frequencies. Modulating a fixed nonzero compactly supported smooth function to arbitrarily high frequency gives a negative form value, also on any fixed nonempty interval. $\square$

There is a second useful decomposition. For $u \neq 0$ define the same-ray and opposite-ray functions

$$G_s(u) = \frac{1}{2 \sinh(|u|/2)}, \quad G_o(u) = \frac{1}{2 \cosh(u/2)}. \quad (\text{S2.5})$$

Then $n_\Gamma(|u|) = \frac{1}{2}(G_s(u) + G_o(u))$. The reflected free network constructed in Section S7 gives G_o . That is a positive and finite kernel, but it is not the even tower by itself. Conversely, the full angular smearing above is an ordinary covariance construction with support on all of S^2 . It is not automatically a state supported in a positive-time half-space, as required for a time-reflection argument. Restricting the angular profile to a hemisphere changes its spectral weights.

The useful conclusion is therefore precise: free defect matter supplies the desired exponents and a regulated positive covariance. The RH target requires the associated difference energy, its fixed local term, its pole term and its arithmetic translations. These are distinct operations on the free data.

S2.1 An auxiliary Gaussian realization of the gamma ratio

The parent endpoint work also gave a useful fixed-space alternative to fractional dimension [13]. It can be understood from elementary Gaussian integration. Retain one complex mode for each $a_n = 2n + 1/2$, and for real p with $a_0 + p > \omega$ define

$$Z_N(p) = \prod_{n=0}^{N-1} \int_{\mathbb{C}} \frac{d^2 z_n}{\pi} e^{-(a_n+p)|z_n|^2} = \prod_{n=0}^{N-1} (a_n + p)^{-1}. \quad (\text{S2.6})$$

For a half-line field $\phi_n(t)$ with action $\int_0^\infty (|\phi'_n|^2 + a_n^2 |\phi_n|^2) dt$ and boundary value z_n , the decaying classical solution is $\phi_n(t) = z_n e^{-a_n t}$. Its action is $a_n |z_n|^2$. Adding the boundary term $p|z_n|^2$ gives the same Gaussian; varying the boundary value gives the Robin condition $(-\partial_t + p)\phi_n(0) = 0$ for the full variational problem. Constants common to the two partition functions cancel in the ratio.

Set $x = (p + 1/2 - \omega)/2$ and $y = x + \omega$. The finite product gives

$$\frac{Z_N(p - \omega)}{Z_N(p + \omega)} = \prod_{n=0}^{N-1} \frac{n + y}{n + x} = \frac{\Gamma(N + y)\Gamma(x)}{\Gamma(y)\Gamma(N + x)}. \quad (\text{S2.7})$$

Since $\Gamma(N + y)/\Gamma(N + x) \sim N^\omega$,

$$\pi^\omega \lim_{N \rightarrow \infty} N^{-\omega} \frac{Z_N(p - \omega)}{Z_N(p + \omega)} = \pi^\omega \frac{\Gamma((p + a)/2)}{\Gamma((p + b)/2)}. \quad (\text{S2.8})$$

Equivalently one can divide by the same finite product at a reference p_0 to cancel the normalization. One real Gaussian mode would give the square root of the ratio, so statistics and multiplicities matter.

This is an auxiliary partition-function model, not a localization of the Wilson network. Angular smearing samples a channel; it does not remove the other angular modes from a partition function. The full scalar sphere has multiplicity $2\ell + 1$, whereas the model retains one copy of each even axial mode. A physical projection or cancellation of the extra modes has not been derived. Here p is a boundary mass, and identifying it with the arithmetic Laplace frequency requires a dynamical dictionary. The Robin response $(a_n + p)^{-1}$, the partition ratio, and a causal transfer are different objects. The supplied factor π^ω , the prime comb, and the rational pole correction remain outside this Gaussian mechanism.

S3 Semicircles, constant drivers, and a restricted rigidity theorem

Choose a plane containing a direction tangent to the defect and its normal, and write $z = x + iy$ with $y > 0$. The common-reference semicircle from $r > 0$ to zero is

$$z_r(\theta) = \frac{r}{2}(1 + \cos \theta + i \sin \theta), \quad 0 \leq \theta < \pi. \quad (\text{S3.1})$$

Inversion preserves the upper half-plane and gives

$$w_r(\theta) = -\frac{1}{z_r(\theta)} = -\frac{1}{r} + \frac{i}{r} \tan(\theta/2). \quad (\text{S3.2})$$

This is a vertical slit. In the chordal convention

$$\partial_t g_t(w) = \frac{2}{g_t(w) - U_t}, \quad g_0(w) = w, \quad (\text{S3.3})$$

the constant driver $U_t = a = -1/r$ has solution

$$g_t(w) = a + \sqrt{(w - a)^2 + 4t}, \quad \gamma_t = a + 2i\sqrt{t}, \quad z_t = -\frac{1}{a + 2i\sqrt{t}}. \quad (\text{S3.4})$$

The square-root branch is fixed by $g_t(w) \sim w$ at infinity. Its expansion is $g_t(w) = w + 2t/w + O(w^{-2})$, so the half-plane capacity is $2t$, consistent with [7, Lecture 2, Proposition 2.17]. Comparing the traces gives the exact dictionary

$$t = \frac{\tan^2(\theta/2)}{4r^2}. \quad (\text{S3.5})$$

The complete semicircle reaches the reference only as $t \rightarrow \infty$. At fixed angle, $r \mapsto \lambda r$ sends $t \mapsto \lambda^{-2}t$. Thus this inverted capacity is neither the coordinate $\log r$ nor the arithmetic shift ω . The formulas identify contours; they do not assert equality of renormalized expectations through an inversion singular at an endpoint.

S3.1 Fixed scalar coupling and fixed special charge

Let the Euclidean Clifford generators satisfy $\{\Gamma_A, \Gamma_B\} = 2\delta_{AB}$. Take one internal direction I and fix a nonzero complex spinor ϵ_c obeying

$$\epsilon_s = 0, \quad J\epsilon_c = -\epsilon_c, \quad J = i\Gamma_I\Gamma_3. \quad (\text{S3.6})$$

Here $\epsilon_s + X^\mu \Gamma_\mu \epsilon_c$ is the relevant conformal Killing spinor, with its constant part set to zero. These are the inherited common bulk conditions. For a regular real curve $X = (x, y) \neq 0$ in the $(1, 3)$ -plane and constant positive scalar arclength coupling, its local line condition is

$$(i\dot{x}\Gamma_1 + i\dot{y}\Gamma_3 + v\Gamma_I)(x\Gamma_1 + y\Gamma_3)\epsilon_c = 0, \quad v = |\dot{X}| > 0. \quad (\text{S3.7})$$

Proposition S3.1 (Rigidity under (S3.6)). *Equation (S3.7) holds if and only if*

$$\frac{\dot{z}}{|\dot{z}|} = \frac{iz^2}{|z|^2}. \quad (\text{S3.8})$$

Consequently $-1/z$ has constant real part and increasing imaginary part. Connected solutions in the upper half-plane are subarcs of circles through zero with center on the real axis, or the degenerate imaginary-axis line. A solution from a nonzero real boundary point to zero is the corresponding oriented upper semicircle.

Proof. The projector implies $\Gamma_I \epsilon_c = -i\Gamma_3 \epsilon_c$. Clifford multiplication reduces (S3.7) to $(A + B\Gamma_1\Gamma_3)\epsilon_c = 0$, where

$$A = x\dot{x} + y\dot{y} + v\dot{y}, \quad B = y\dot{x} - x\dot{y} + vx. \quad (\text{S3.9})$$

Since $(\Gamma_1\Gamma_3)^2 = -1$, multiplication by $A - B\Gamma_1\Gamma_3$ gives $(A^2 + B^2)\epsilon_c = 0$. The coefficients are real and the spinor is nonzero, so $A = B = 0$. Solving gives

$$\frac{\dot{x}}{v} = -\frac{2xy}{x^2 + y^2}, \quad \frac{\dot{y}}{v} = \frac{x^2 - y^2}{x^2 + y^2}, \quad (\text{S3.10})$$

which is (S3.8). Conversely it makes $A = B = 0$. Moreover $d(-1/z)/ds = iv/|z|^2$. The level set $\text{Re}(-1/z) = a \neq 0$ is $x^2 + y^2 = -x/a$, and $a = 0$ gives the imaginary axis. $\square$

For instance, an inverted deformation $w(s) = a + \varepsilon h(s) + is$ preserves these conditions only if $\varepsilon h'(s) = 0$. A constant h changes the slit location; a nontrivial fixed-endpoint deformation fails. A regular capacity-parametrized Loewner trace satisfying the same conditions therefore has constant driving.

This is a bulk theorem for one plane, one charge and one scalar prescription. It is not a classification of supersymmetric contours with varying scalar directions, other conformal Killing spinors, additional endpoint degrees of freedom, or nonplanar embeddings. Section S5 supplies a constructive control showing that a changed scalar prescription allows arbitrary tangents.

S4 Smooth Wilson variation and the Loewner insertion

S4.1 Conventions and the exact variation

Work with smooth finite matrix fields and a regular C^2 curve $X : [0, 1] \rightarrow \mathbb{R}^4$. Let A_μ and the scalar fields X_I be Hermitian, and let $n^I(s)$ be a unit internal vector. Define

$$\mathcal{M}(s) = iA_\mu(X(s))\dot{X}^\mu(s) + \Phi(X(s), s)v(s), \quad \Phi = n^I X_I, \quad v = |\dot{X}|. \quad (\text{S4.1})$$

The ordered transport $U(b, a)$ solves $\partial_b U(b, a) = \mathcal{M}(b)U(b, a)$, $U(a, a) = I$; larger parameters stand to the left. Our curvature and covariant derivative are

$$F_{\nu\mu} = \partial_\nu A_\mu - \partial_\mu A_\nu - i[A_\nu, A_\mu], \quad D_\nu \Phi = \partial_\nu \Phi - i[A_\nu, \Phi]. \quad (\text{S4.2})$$

Vary the curve by $\eta = \delta X$ and the internal vector by δn , holding the spacetime fields fixed.

Proposition S4.1 (Ordered transport variation). *With $C_s = iA_\nu(X(s))\eta^\nu(s)$,*

$$\delta U(1, 0) = C_1 U(1, 0) - U(1, 0)C_0 + \int_0^1 U(1, s)\mathcal{I}(s)U(s, 0) ds, \quad (\text{S4.3})$$

$$\mathcal{I} = iF_{\nu\mu}\eta^\nu \dot{X}^\mu + vD_\nu \Phi \eta^\nu + \Phi \frac{\dot{X} \cdot \dot{\eta}}{v} + vX_I \delta n^I. \quad (\text{S4.4})$$

Proof. Differentiating the defining ODE gives $\delta U = \int U \delta \mathcal{M} U$. Direct differentiation of $\mathcal{M}$ yields $\delta \mathcal{M} = \dot{C} + [C, \mathcal{M}] + \mathcal{I}$. Meanwhile

$$\frac{d}{ds} \{U(1, s)C_s U(s, 0)\} = U(1, s)(\dot{C} + [C, \mathcal{M}])U(s, 0). \quad (\text{S4.5})$$

Integration proves the formula, including its ordering and boundary signs. $\square$

For constant n and fixed endpoints, integrate the arclength term by parts. Writing $T = dX/d\ell$, $k = dT/d\ell$, one obtains

$$\begin{aligned}\delta U &= \int_C U(1, s) \eta^\nu \mathcal{D}_\nu U(s, 0) d\ell, \\ \mathcal{D}_\nu &= iF_{\nu\mu} T^\mu + (\delta_\nu^\mu - T_\nu T^\mu) D_\mu \Phi - \Phi k_\nu.\end{aligned}\tag{S4.6}$$

In particular $T^\nu \mathcal{D}_\nu = 0$, as required for a fixed-endpoint reparametrization. The curvature term is necessary because the scalar couples to arclength. If n varies, one must retain its additional terms in (S4.4).

Let the endpoints be $x = X(0)$ and $y = X(1)$, both on the defect, and consider $\mathcal{O} = \bar{q}(y)U(1, 0)q(x)$. The boundary terms in (S4.3) combine with endpoint differentiation to give

$$\begin{aligned}\delta \mathcal{O} &= \eta_y^\nu (\partial_\nu \bar{q} + i\bar{q}A_\nu)(y)Uq(x) + \bar{q}(y)U\eta_x^\nu (D_\nu q)(x) \\ &\quad + \int_0^1 \bar{q}(y)U(1, s)\mathcal{I}(s)U(s, 0)q(x) ds.\end{aligned}\tag{S4.7}$$

Here $D_\nu q = \partial_\nu q - iA_\nu q$, and endpoint displacements are tangent to the defect. Changing endpoint R polarizations contributes further terms. A defect field q is not defined at a bulk tip.

S4.2 Regular inverse-map deformations

Let $f_t = g_t^{-1}$ for (S3.3), and choose a fixed smooth reference contour $\zeta(s)$ in a region where f_t is regular and $\zeta(s) \neq U_t$. Differentiating $g_t(f_t(w)) = w$ gives

$$X_t(s) = f_t(\zeta(s)), \quad \eta_t(s) = \partial_t X_t(s) = -\frac{2f_t'(\zeta(s))}{\zeta(s) - U_t}.\tag{S4.8}$$

Insert its real and imaginary components into (S4.4) and (S4.7). This is a concrete field-dependent evolution of the transport on a pulled-back test curve. It is not a variation formula for the singular growing tip. Regular real boundary segments remain on the defect in this planar interpretation. A growing prefix instead carries an uncontracted color index at its interior endpoint; it requires a specified tip state or a conditional completion of the curve.

If $dU_t = \sqrt{\kappa} dB_t$ for standard Brownian motion B_t , then at fixed w the inverse-map equation is of finite variation until a singularity is reached. In centered coordinates $\mathfrak{f}_t(w) = f_t(w + U_t)$, stopped smooth Ito calculus gives

$$d\mathfrak{f}_t(w) = \left[-\frac{2\mathfrak{f}_t'(w)}{w} + \frac{\kappa}{2}\mathfrak{f}_t''(w) \right] dt + \sqrt{\kappa}\mathfrak{f}_t'(w) dB_t.\tag{S4.9}$$

This is only a map identity. No scalar arclength integral or quantum Wilson transport on an unregulated stochastic trace is defined by it.

S4.3 A precise obstruction to geometry-only closure

Proposition S4.2. *There is no universal multiplicative shape generator for a single Wilson transport whose coefficient depends only on the curve and its deformation, independently of the background fields.*

Proof. In an abelian theory take $A_x = cy$, $A_y = \Phi = 0$ and the straight path $X(s) = (s, 0)$, $0 \leq s \leq 1$. Then $U = 1$ for every real c . For $\eta = (0, h(s))$ with $h(0) = h(1) = 0$,

$$\delta U = ic \int_0^1 h(s) ds. \quad (\text{S4.10})$$

The same curve, deformation and original transport thus have different derivatives. As a positive control, a pure gauge $A = d\chi$ has fixed-endpoint transport independent of the contour. $\square$

The proposition does not exclude closure of an averaged correlation function after field equations, Ward identities or a protected reduction are supplied. It identifies the extra information such a closure must use. At leading free endpoint order $U = I$, fixed-endpoint interior deformations do not change (S2.3); the nontrivial information must enter at higher order or through another specified sector.

S4.4 The tangent connection as a one-form

For a constant real isometry $M : \mathbb{R}^4 \rightarrow \mathbb{R}^6$, set $n = MT$, so that $\Phi|dX| = \Phi_\mu dX^\mu$ with $\Phi_\mu = M^A{}_\mu X_A$. Define $\mathcal{B}_\mu = iA_\mu + \Phi_\mu$. Then the fixed-endpoint variation takes the ordinary one-form form

$$\begin{aligned} \delta U_M &= \int U_M(1, s) \mathcal{F}_{\nu\mu} \eta^\nu \dot{X}^\mu U_M(s, 0) ds, \\ \mathcal{F}_{\nu\mu} &= \partial_\nu \mathcal{B}_\mu - \partial_\mu \mathcal{B}_\nu - [\mathcal{B}_\nu, \mathcal{B}_\mu] \\ &= iF_{\nu\mu} + D_\nu \Phi_\mu - D_\mu \Phi_\nu - [\Phi_\nu, \Phi_\mu]. \end{aligned} \quad (\text{S4.11})$$

For moving endpoints the boundary insertion is $\mathcal{B}_\nu \eta^\nu$. These signs follow from the same ordered ODE. Supersymmetry of the line is not an assertion that this complex curvature vanishes on arbitrary field configurations.

S5 Tangent coupling with the full chiral defect projector

S5.1 Field content and conventions

The defect is $x^3 = 0$ in Euclidean spacetime (x^0, x^1, x^2, x^3) . The bulk has a gauge field A_μ , six adjoint scalar fields X_A , $A = 4, \dots, 9$, and their supersymmetric fermion partners. The triplets (X_4, X_5, X_6) and (X_7, X_8, X_9) transform under $SU(2)_H$ and $SU(2)_V$, respectively. Defect matter includes fundamental scalars q_m , $m = 1, 2$, forming an H doublet, and fermions forming a V doublet; $\bar{q}_m$ are the physical antifundamental conjugates. These are the bulk/defect assignments of [4, 6].

Work in a 32-dimensional complex Euclidean Clifford module, with

$$\chi = -i\Gamma_0\Gamma_1 \cdots \Gamma_9, \quad D = \Gamma_3\Gamma_4\Gamma_5\Gamma_6, \quad \chi\epsilon = \epsilon, \quad D\epsilon = \epsilon. \quad (\text{S5.1})$$

The second equation is the unfluxed D3–D5 intersection condition. It is the projector appearing in [5, eqs. (2.7), (4.1)–(4.3), (4.9)]; the terminating-brane boundary conditions and Nahm pole are not part of our assumptions. The displayed conventions fix the chirality and orientation signs. Complex ranks below are not counts of real Euclidean Majorana–Weyl supercharges.

S5.2 A normal plane

For the (x^1, x^3) -plane take the connection

$$\mathcal{B} = iA_1 dx^1 + iA_3 dx^3 + X_8 dx^1 + X_6 dx^3. \quad (\text{S5.2})$$

This is the tangent prescription $n = (\dot{x}^1 e_8 + \dot{x}^3 e_6)/|\dot{x}|$, so the scalar direction changes with orientation. In the usual local supersymmetry variation of scalar-coupled transport, a constant Poincare parameter is annihilated by every tangent if

$$[\dot{x}^1(i\Gamma_1 + \Gamma_8) + \dot{x}^3(i\Gamma_3 + \Gamma_6)]\epsilon = 0. \quad (\text{S5.3})$$

This is the tangent construction of [3], now with (S5.1) imposed as well. Define

$$K_1 = i\Gamma_8\Gamma_1, \quad K_n = i\Gamma_6\Gamma_3. \quad (\text{S5.4})$$

Proposition S5.1 (Planar charge space). *The common subspace $\chi = D = +1$, $K_1 = K_n = -1$ has complex dimension two. Every parameter in it obeys (S5.3) for every smooth planar tangent.*

Proof. The four displayed matrices are commuting Hermitian involutions. Their orthogonal projector is

$$P_{\text{plane}} = \frac{1}{16}(1 + \chi)(1 + D)(1 - K_1)(1 - K_n). \quad (\text{S5.5})$$

Every nonidentity Clifford word in its expansion has zero trace, and no nonempty product of these four generators is scalar. Hence $\text{tr } P_{\text{plane}} = 32/16 = 2$. Multiplication of $K_1\epsilon = -\epsilon$ by Γ_8 , and of $K_n\epsilon = -\epsilon$ by Γ_6 , gives the two linear equations in (S5.3) separately. $\square$

The triplet assignments are necessary within the stated real, constant-map ansatz. The defect projector D commutes with a defect tangent gamma and with a V gamma; it anticommutes with the normal gamma and with an H gamma. Pairing a tangent with H, or the normal with V, therefore gives an involution anticommuting with D and no simultaneous eigenspinor. More generally, split a proposed real scalar vector into H and V components. For an independently variable defect tangent its wrong H component lies in the opposite D -eigenspace from the remaining line equation, so it must annihilate ϵ separately. Its Clifford square is its squared real length, making it invertible unless that component is zero. The normal argument exchanges H and V. Complex null internal vectors and couplings restricted to one constrained tangent are outside this argument. Proposition S5.1 is the nonzero control.

S5.3 One charge across all defect directions

To accommodate an angular family, introduce three tangent conditions

$$K_\mu^{(s)} = is_\mu \Gamma_{7+\mu} \Gamma_\mu, \quad \mu = 0, 1, 2, \quad s_\mu \in \{+1, -1\}. \quad (\text{S5.6})$$

The corresponding oriented tangent isometry has determinant $s_0 s_1 s_2$. Direct Clifford multiplication gives

$$K_0^{(s)} K_1^{(s)} K_2^{(s)} = (s_0 s_1 s_2) \chi D. \quad (\text{S5.7})$$

Proposition S5.2 (Orientation and the all-direction charge). *For chirality $\chi = c$ and defect sign $D = d$, all three tangent eigenvalues can be -1 , together with $K_n = -1$, if and only if the tangent isometry has determinant $-cd$. With that sign the common space has complex dimension one; with the opposite sign it is zero.*

Proof. Taking eigenvalues in (S5.7) proves necessity. For the allowed sign, one tangent condition follows from the other two and χ, D . The remaining five commuting conditions $\chi = c$, $D = d$, $K_n = -1$, $K_0^{(s)} = K_1^{(s)} = -1$ are independent. Their projector has trace $32/2^5 = 1$, by the same nonidentity-word trace argument as above. Orthogonal changes of the oriented tangent and internal frames give the general isometry statement. $\square$

For $c = d = +1$, choose once and for all

$$Me_0 = -e_7, \quad Me_1 = e_8, \quad Me_2 = e_9, \quad Me_3 = e_6. \quad (\text{S5.8})$$

Then $M^T M = I$ and the connection is

$$\mathcal{B} = iA_\mu dx^\mu - X_7 dx^0 + X_8 dx^1 + X_9 dx^2 + X_6 dx^3, \quad \mu = 0, 1, 2, 3. \quad (\text{S5.9})$$

The same nonzero constant complex parameter works for every Euclidean four-dimensional tangent, hence for all normal planes indexed by S^2 . This is not a statement about arbitrary real Lorentzian trajectories with real internal couplings.

S5.4 The remaining H weight

Set $T_H = i\Gamma_4\Gamma_5$. The exact identity

$$D = K_n T_H \quad (\text{S5.10})$$

implies $T_H \epsilon = -\epsilon$ for our signs. The tangent projectors reduce the spacetime/V factors but do not introduce the other H weight. Thus a surviving constant charge has the same one-dimensional H factor at both endpoints. Using Baker's generator convention, the plus-normal combination $iA_3 + X_6$ is labeled by preserved generators Q_{i2} ; the conjugate convention exchanges 1 and 2. Spinor-parameter and generator indices are related by charge conjugation, so the endpoint labels must be fixed from the physical transformations, not read off from a gamma eigenvalue alone.

S6 Endpoint polarization, vanishing, and the adjoint

S6.1 The same-charge endpoint calculation

The physical scalar variations may be written $\delta q_m = 2\bar{\zeta}_{im}\Psi_i$ and $\delta \bar{q}_m = 2\bar{\Psi}_i\zeta_{im}$, with the related physical generator conjugation; see [6, Appendix C]. Complexification does not make the actions on q and $\bar{q}$ independently adjustable. For the plus-normal choice the annihilated scalar components are q_2 and $\bar{q}_1$.

Here is a basis-independent polarization version of that statement. Suppress the nonzero spacetime/V factor of the charge, let its H factor be $a \in \mathbb{C}^2 \setminus \{0\}$, and write

$$q(v) = \sum_m v_m q_m, \quad \bar{q}(w) = \sum_m w_m \bar{q}_m, \quad E = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \quad (\text{S6.1})$$

In the generator convention just specified, the invariant tensors give

$$Q(a)q(v) \propto (a^T E v)\Psi, \quad Q(a)\bar{q}(w) \propto (a^T w)\tilde{\Psi}. \quad (\text{S6.2})$$

This follows by rotating the representative conditions $Q_{i2}q_2 = Q_{i2}\bar{q}_1 = 0$, with E the fundamental antisymmetric tensor and the second contraction the fundamental/dual evaluation. The suppressed fermion components are arbitrary, so an operator annihilator requires the scalar coefficient to vanish.

Proposition S6.1 (Zero same-charge scalar pairing). *For the constant fixed-H-factor charge above, nonzero ordinary fundamental and antifundamental scalar endpoints separately killed by that charge have zero mutual free contraction. They are not a physical adjoint pair.*

Proof. The two nullspaces in (S6.2) are

$$v = \lambda a, \quad w = \mu E a. \quad (\text{S6.3})$$

The H tensor of the free propagator is δ_{mn} , so its polarization factor is $v^T w = \lambda \mu a^T E a = 0$. Physical conjugation instead requires $w = v^*$ and gives $v^T v^* = \|v\|^2 > 0$. If $v = \lambda a \neq 0$, then $a^T v^* = \lambda^* \|a\|^2 \neq 0$, so the physical conjugate is not killed by that same charge. $\square$

In the representative basis, q_2 pairs nontrivially with $\bar{q}_2$, whereas its same-charge partner is $\bar{q}_1$. The physical adjoint pair is a useful nonzero control, but its dressed expectation still requires an independent positivity argument. At distinct endpoints the two supersymmetry variations contain independent fermion fields, so they cannot cancel each other as bare operator identities. In this ansatz the bulk line variation already vanishes separately. Junction matter or a fermionic superconnection would change the operator and require a new calculation.

S6.2 The full bilinear has nonzero R charge

The fields in (S5.2) and (S5.9) are neutral under the H rotation fixing X_6 . Assign q_1, q_2 weights $+1/2, -1/2$. Then $\bar{q}_1$ also has weight $-1/2$ and

$$\mathcal{O}_C = \bar{q}_1(b) U_M(C) q_2(a) \quad (\text{S6.4})$$

has charge -1 . If the vacuum and the definition of the operator preserve this symmetry, its expectation obeys

$$\langle \mathcal{O}_C \rangle = e^{-i\alpha} \langle \mathcal{O}_C \rangle \quad \text{for every } \alpha, \quad \langle \mathcal{O}_C \rangle = 0. \quad (\text{S6.5})$$

This selection rule extends beyond the free contraction. It does not construct an ultraviolet-finite line operator. A compensating charged insertion, a different endpoint sector or a symmetry-breaking vacuum changes its assumptions. A fixed-endpoint deformation of the same neutral connection carries no compensating charge.

S6.3 Why the original semicircle is a nonzero control

The restriction is to a constant Poincare charge and the displayed tangent map. On (S3.1), that map has scalar direction $n_8 = -\sin \theta$, $n_6 = \cos \theta$: the scalar normal direction changes sign along with the tangent. The common Poincare charge therefore keeps its H weight fixed.

The original semicircle with constant scalar arclength coupling instead uses a position-dependent conformal Killing spinor. Its local endpoint conditions can select different H factors at the two ends. Baker's same-polarization semicircle has a nonzero leading endpoint contraction, proportional to $g^2/(4\pi|a-b|)$ in that paper's normalization [6, Section 3.2]. It is consequently a successful control outside Proposition S6.1, rather than a counterexample to it. This distinction is also why the rigidity theorem and tangent-map construction must not be conflated.

S6.4 The adjoint changes the scalar map

For a tangent-coupled line and Hermitian matrix fields,

$$U_M(C)^\dagger = U_{-M}(C^{-1}), \quad U_M(C^{-1}) = U_M(C)^{-1}. \quad (\text{S6.6})$$

To prove the first identity, reverse the product order in the discretized ordered exponential and conjugate each exponent. The gauge term changes sign under conjugation; after reversing the contour it is again $iA_\mu dx^\mu$, while the scalar one-form has acquired $-M$. The second identity is ordinary inverse transport. Thus contour reversal with unchanged tangent map gives the inverse, not the Hermitian adjoint. A reflected norm must implement the former identity with the appropriate spacetime pullback. This is the purpose of the next section.

S7 Physical reflection and the reference color junction

S7.1 Ordinary time reflection

Reflect a time direction tangent to the defect:

$$\vartheta x = Rx, \quad R = \text{diag}(-1, 1, 1, 1), \quad b = 0 \in \{x^0 = 0\} \cap \{x^3 = 0\}. \quad (\text{S7.1})$$

The positive half-space is $x^0 > 0$. In the bosonic adjoint convention, A transforms as a one-form, all six Hermitian scalars are even under geometric time reflection, and conjugation exchanges q_m with $\bar{q}_m$. Fermionic reflection and an interacting regulator are additional requirements, not consequences of these bosonic rules.

Let $C : a \rightarrow b$ lie in the positive half-space except at b , with a on the defect. Use M from (S5.8). Pullback of the fields first gives $U_{MR}(\vartheta C)$; applying (S6.6) then gives

$$\Theta_0 U_M(C) = U_N(\vartheta C^{-1}), \quad N = -MR. \quad (\text{S7.2})$$

Explicitly,

$$Ne_0 = -e_7, \quad Ne_1 = -e_8, \quad Ne_2 = -e_9, \quad Ne_3 = -e_6. \quad (\text{S7.3})$$

The time component retains its scalar map and the spatial components reverse it. This ordering calculation agrees with the distinction between geometric reflection and scalar-coupled adjoint transport in [10, eqs. (4)–(7)].

S7.2 The actual paired operator and the conditional Gram statement

Define a covariant color vector at the reference by

$$F_i^\alpha = [U_M(C_i)]^\alpha{}_\beta q_2^\beta(a_i). \quad (\text{S7.4})$$

Its reflected row is $\bar{q}_{2,\gamma}(\vartheta a_i)[U_N(\vartheta C_i^{-1})]^\gamma{}_\alpha$. The proposed kernel is

$$K_{ij} = \left\langle \bar{q}_{2,\gamma}(\vartheta a_i)[U_N(\vartheta C_i^{-1})]^\gamma{}_\alpha [U_M(C_j)]^\alpha{}_\beta q_2^\beta(a_j) \right\rangle. \quad (\text{S7.5})$$

There is an identity color contraction at b , not another defect field there. Under a gauge transformation, $F_i \mapsto g(b)F_i$ and its reflected row transforms by $g(b)^{-1}$, so the paired observable is gauge invariant. The individual vector is not a gauge-invariant state without a specified reference charge sector.

Assumption S7.1. A regulator and reference color sector admit Osterwalder–Schrader positivity for these half-space functionals. The regulated endpoint and junction operators belong to that domain, and their limiting renormalization preserves the inequality whenever a limit is claimed.

Proposition S7.2 (Conditional positivity). *Under the regulated part of Assumption S7.1, the matrix in (S7.5) is positive semidefinite for every finite collection of admissible contours and endpoints.*

Proof. Apply the assumed OS inequality to $\sum_i c_i F_i$, including its reference color fiber, to obtain $\sum_{ij} \bar{c}_i K_{ij} c_j \geq 0$. For a concrete bosonic sufficient model, suppose the two half-integrals factor over boundary data B with positive measure ν . They give vector wavefunctions $\psi_i^\alpha(B)$ and

$$K_{ij} = \int \sum_{\alpha} \overline{\psi_i^\alpha(B)} \psi_j^\alpha(B) d\nu(B), \quad \nu \geq 0. \quad (\text{S7.6})$$

The claimed inequality is then the integral of a squared norm. Boundary gauge transformations act unitarily on the displayed fiber. $\square$

This conditional statement locates the missing physical input. A positive bosonic factorization is not asserted for the full fermionic $\mathcal{N} = 4$ defect measure. Gauge invariance of (S7.5) alone proves no reflection inequality. Gauss law, the reference charge sector, the regulator, and junction counterterms must all be addressed in an interacting realization.

S7.3 An unconditional free kernel

At leading free endpoint order $U = I$. Strip the positive color and coupling constants and write $a_i = (\tau_i, \mathbf{z}_i, 0)$ with $\tau_i > 0$, $\mathbf{z}_i \in \mathbb{R}^2$. Then

$$K^{(0)}(a_i, a_j) = \frac{1}{\sqrt{(\tau_i + \tau_j)^2 + |\mathbf{z}_i - \mathbf{z}_j|^2}}. \quad (\text{S7.7})$$

Its representation

$$K^{(0)}(a_i, a_j) = 4\pi \int_{\mathbb{R}^2} \frac{d^2 k}{(2\pi)^2} \frac{e^{-|k|(\tau_i + \tau_j)} e^{ik \cdot (\mathbf{z}_j - \mathbf{z}_i)}}{2|k|} \quad (\text{S7.8})$$

is an integral of products $\overline{\psi_i(k)} \psi_j(k)$ with positive measure. This proves free positivity independently of Assumption S7.1. On the time ray $a_r = (r, 0, 0, 0)$ it reduces to

$$K^{(0)}(r, s) = \frac{1}{r + s} = \int_0^\infty e^{-\lambda r} e^{-\lambda s} d\lambda, \quad \sqrt{rs} K^{(0)}(r, s) = G_o(\log r - \log s). \quad (\text{S7.9})$$

The diagonal is finite for $r > 0$. The kernel is Hankel in r, s , and translation invariant after the radial weights and logarithmic coordinates. It is the opposite-ray kernel of (S2.5), not the full even archimedean tower. The full-sphere profile in Section S2 cannot be used as the same half-space state without changing its support and recomputing its weights.

S7.4 The two branches preserve different constant charges

Let $\widetilde{K}_\mu = i\Gamma_{Me_\mu} \Gamma_\mu$ for $\mu = 0, 1, 2$, and $K_n = i\Gamma_6 \Gamma_3$. With $\chi = D = +1$, the two branches require the following eigenvalues:

Involution	Ket with M	Reflected bra with N
$\widetilde{K}_0$	-1	-1
$\widetilde{K}_1, \widetilde{K}_2$	-1	$+1$
K_n	-1	$+1$

Each all-direction family has one complex compatible charge; in the (x^0, x^3) -plane each has two. Their common subspace is zero when time and normal tangents are independently required, because the normal conditions are opposite. The normal projectors also select opposite H weights: the ket uses q_2 and the physical bra uses $\bar{q}_2$ with the conjugate-weight charge. The identity tensor at b cannot compensate a nonzero variation on a branch.

As a bulk control, a straight time-directed contour needs only $\widetilde{K}_0 = -1$. That condition is common to the maps and gives a four-dimensional complex chiral defect solution space. It is a bulk control, not a statement that a constant charge annihilates the physical adjoint pair of endpoint scalars.

S7.5 Internal rotations do not automatically preserve positivity

The proper rotations by π about H axis 4 and V axis 7 are

$$S_H = \text{diag}(1, -1, -1), \quad S_V = \text{diag}(1, -1, -1). \quad (\text{S7.10})$$

For $S = S_H \oplus S_V$, one checks $SN = M$. Thus an internally rotated reflection aligns the two bulk scalar maps. But S_H acts on the endpoint doublet by $\pm i\sigma_1$, up to basis convention. It takes $\bar{q}_2$ to a phase times $\bar{q}_1$, restoring the zero diagonal contraction of Proposition S6.1.

On the full free endpoint doublet the inserted metric is not positive. Without a phase adjustment its eigenvalues are $\pm i$; with a phase making it Hermitian they are $+1, -1$. For instance

$$(1, -1)\sigma_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} / 2 = -1. \quad (\text{S7.11})$$

No overall phase makes both eigenvalues positive. The identity metric from ordinary adjoint pairing is the control. More generally, if a unitary S on a positive Hilbert space satisfies $\langle \psi, S\psi \rangle \geq 0$ for every ψ , then S is positive self-adjoint; unitarity forces its spectrum to be $\{1\}$ and $S = I$. Projection to one rotation eigenspace is a different operation and here mixes the fixed H endpoint weights.

This is a counterexample for the displayed doublet state space. It does not contradict the special large-rectangle rotated-reflection conjecture distinguished from ordinary reflection in [10, eqs. (11)–(16)].

S7.6 A free test of an odd-H normal reflection

A different proposal in the parent investigation reflected across the defect and assigned odd parity to the H scalars. For the canonical free scalar algebra, a real mode of frequency $\nu > 0$ at distance $a > 0$ from a reflection plane has ordinary norm $e^{-2\nu a}/(2\nu) > 0$; assigning it odd reflection parity changes the sign. The failure is also visible in a gauge-invariant observable. For a non-Abelian group with $\text{tr}(T_a T_b) = \delta_{ab}/2$, take the Hermitian Wick polynomial

$$\mathcal{O} = i \text{tr}(H[V_1, V_2]), \quad (\text{S7.12})$$

where H is one H scalar and V_1, V_2 are distinct V scalars. Writing $[T_b, T_c] = if_{bca}T_a$, its coefficient is $-f_{bca}/2$. Distinct species have only their matching free contractions, hence at separated reflected points

$$\langle \Theta_{\text{even}} \mathcal{O}(x) \mathcal{O}(x) \rangle = \frac{1}{4} \sum_{abc} f_{abc}^2 D_0(\vartheta x - x)^3 > 0, \quad (\text{S7.13})$$

with D_0 the positive scalar covariance. Odd H and even V parity give the negative of this value. This excludes that reflection on the unrestricted canonical free bulk observable algebra. It does not classify all interacting normal-reflection actions or restricted observable sectors. The even assignment supplies the positive control.

S8 A smooth reference join and a direct bulk response

S8.1 A contour family that meets the reflection plane smoothly

Let v be the incoming unit ket tangent at b . Its reflected reverse has outgoing tangent $-Rv$. A smooth join requires

$$v = -Rv, \quad (\text{S8.1})$$

so v is parallel to the time direction e_0 . The internal scalar directions then agree as well, since $N(-Rv) = Mv$. For a kernel involving many contours, every ket should have the same reference tangent. The old semicircles arrive along $-e_3$ and return along $+e_3$ under time reflection, producing a backtracking join instead.

Contour freedom provides a simple replacement. In the (x^0, x^3) -plane set

$$X_{T,\eta}(t) = (Tt, \eta T f(t)), \quad f(t) = t^2(1-t), \quad 0 \leq t \leq 1, \quad T > 0, \quad \eta \geq 0. \quad (\text{S8.2})$$

The ket traverses the curve from $t = 1$ to zero. The endpoints are on the defect, its interior is in positive time and, for $\eta > 0$, strictly above the defect. Since $f'(0) = 0$, its incoming unit tangent is $-e_0$. The reflected return is regular with a continuous scalar direction even when the two branches have different T or η . Curvature need not agree.

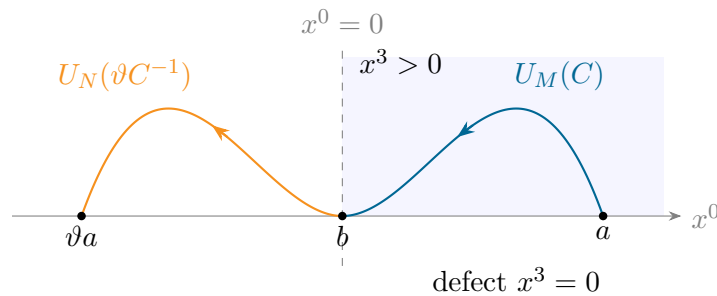


Figure S1: A polynomial bulge and its ordinary reflected adjoint. The ket runs from a to b , and the bra from b to ϑa . Both branches lie above the defect and meet with tangent $-e_0$. The scalar maps differ away from the join. Vertical scale is enlarged.

This construction removes a fixed-angle geometric and scalar-direction cusp in the direct bulk term. It does not prove absence of defect contact or junction counterterms. No driver or capacity law for (S8.2) is computed here; it is a smooth contour control for the field-theory problem, not an identification with ω .

S8.2 The specified Gaussian perturbative sector

Use the free Feynman-gauge propagators

$$\langle A_\mu^a(x) A_\nu^b(y) \rangle_0 = \frac{g^2 \delta^{ab} \delta_{\mu\nu}}{4\pi^2 |x - y|^2}, \quad \langle X_A^a(x) X_B^b(y) \rangle_0 = \frac{g^2 \delta^{ab} \delta_{AB}}{4\pi^2 |x - y|^2}. \quad (\text{S8.3})$$

This defines the direct exchange of a bulk field between the two line branches. Endpoint self-energy, endpoint-to-line interactions, defect vertices and counterterms are excluded from this sector. Only their consistent sum could be a gauge-independent interacting answer.

On a single branch the gauge/scalar numerator cancels by $M^T M = I$ or $N^T N = I$. If v_i, v_j denote oriented original ket tangents, the cross-branch numerator instead is

$$\begin{aligned} -(-Rv_i) \cdot v_j + [N(-Rv_i)] \cdot Mv_j &= v_i \cdot (R + I)v_j \\ &= 2v_{i,\text{sp}} \cdot v_{j,\text{sp}}. \end{aligned} \quad (\text{S8.4})$$

Here sp consists of the three directions unchanged by time reflection, including the defect normal. Near a regular reference join satisfying (S8.1), the spatial tangents vanish linearly in arclength. The numerator is $O(st)$ and the denominator is comparable to $(s + t)^2$. The local double integral is finite. This local estimate concerns (S8.3), not all possible defect graphs.

For the self-pairing of (S8.2), T cancels by scale invariance and the stripped cross term is

$$I(\eta) = 2\eta^2 \int_0^1 \int_0^1 \frac{f'(s)f'(t)}{(s+t)^2 + \eta^2[f(s) - f(t)]^2} ds dt. \quad (\text{S8.5})$$

If $\sum_a T_a T_a = C_F I$ in the endpoint representation, its contribution relative to the free endpoint propagator is $g^2 C_F I(\eta)/(4\pi^2)$. There is no factor $1/2$ for an exchange between two distinct ordered branches.

Proposition S8.1 (The direct bulk bulge coefficient). *For $f(t) = t^2(1 - t)$,*

$$I(\eta) = \left(\frac{3}{2} - 2 \log 2\right) \eta^2 + O(\eta^4), \quad \frac{3}{2} - 2 \log 2 = 0.1137056388801094 \dots > 0. \quad (\text{S8.6})$$

One may bound the absolute remainder by $2\eta^4$ for real η .

Proof. The quadratic coefficient is $2A$, where

$$A = \int_0^1 \int_0^1 \frac{(2s - 3s^2)(2t - 3t^2)}{(s + t)^2} ds dt. \quad (\text{S8.7})$$

Put $r = s/(s + t)$ and $u = s + t$. The Jacobian is u . By symmetry, integrate $0 \leq r \leq 1/2$ and $0 \leq u \leq 1/(1 - r)$ and multiply by two. The integrand including the Jacobian is $r(1 - r)[4 - 6u + 9r(1 - r)u^2]u$. Its u integral, including that factor two, simplifies to $r^2/[2(1 - r)^2]$. Hence

$$A = \frac{1}{2} \int_0^{1/2} \frac{r^2}{(1 - r)^2} dr = \frac{3}{4} - \log 2. \quad (\text{S8.8})$$

For the remainder, $|f'| \leq 1$, $|f'(s)| \leq 2s$ and $|f(s) - f(t)| \leq |s - t| \leq s + t$. Subtracting the denominator at $\eta = 0$ bounds the absolute integrand difference by

$$2\eta^4 \frac{|f'(s)f'(t)|[f(s) - f(t)]^2}{(s + t)^4} \leq 8\eta^4 \frac{st}{(s + t)^2} \leq 2\eta^4. \quad (\text{S8.9})$$

The integration square has area one. This justifies the expansion uniformly even at the apparent singular corner. $\square$

Numerical controls give $I(0) = 0$, $I(0.3) \simeq 0.0102220294386856$ and $I(0.7) \simeq 0.0553790171218514$. These check the integral and its small-bulge coefficient; they are not estimates of the complete interacting response.

There is also a positive Gaussian interpretation of this cross term. The reflected four-dimensional scalar covariance has representation

$$D_0(\vartheta x - y) = \int_{\mathbb{R}^3} \frac{d^3 k}{(2\pi)^3} \frac{e^{-|k|(x^0+y^0)} e^{ik \cdot (\mathbf{y} - \mathbf{x})}}{2|k|}. \quad (\text{S8.10})$$

Smear it against the three spatial tangent currents of each ket and sum the components as in (S8.4). This is a Gram matrix. The vanishing of those currents at b permits the finite join limit just estimated. This proves positivity of the displayed Gaussian current sector, without extending it to the omitted diagrams.

S8.3 Why a positive norm need not be protected

For an actual reflected pairing a smooth deformation has the structure

$$\delta K_{ij} = \langle \Theta_0(\delta F_i) F_j \rangle + \langle \Theta_0 F_i \delta F_j \rangle, \quad (\text{S8.11})$$

with the insertions of Section S4 and any renormalized contact terms. Even if a deformation of a ket is Q -exact, $\delta\psi = Q\lambda$, and $Q\psi = 0$, its physical norm obeys

$$\delta \|\psi\|^2 = 2 \operatorname{Re} \langle \psi, Q\lambda \rangle = 2 \operatorname{Re} \langle Q^\dagger \psi, \lambda \rangle. \quad (\text{S8.12})$$

The first condition does not imply $Q^\dagger \psi = 0$. For example, on a Hilbert space with orthonormal e_1, e_2 , let $Qe_2 = e_1$, $Qe_1 = 0$, $\psi = e_1$, $\lambda = e_2$. Then $Q^2 = 0$, $Q\psi = 0$, but (S8.12) equals two. Adding a vector e_3 killed by both Q and $Q^\dagger$ supplies a zero-first-variation control. Thus a common-charge Ward identity and a physical positive norm are different requirements.

S8.4 The complete response still required

The immediate continuation is the complete first interaction-order shape response of (S7.5), beginning with a positive bulge $\eta_0 > 0$ in (S8.2). Fix the gauge, endpoint normalization, reference color sector and renormalization prescription before comparing deformations. Alongside the direct bulk exchange, include endpoint-to-line gauge exchange, the defect H-scalar derivative coupling, endpoint self-energy, and all defect and junction counterterms at that order. The interaction vertices must be taken from a consistent defect action, such as the conventions used in [4, 6]. Endpoint renormalization may cancel in a specified ratio, but that cancellation must be shown.

The flat limit needs separate analysis. At $\eta = 0$ the entire contour lies on the defect; for $\eta > 0$ only its endpoints do. Defect contact terms can therefore make the full small-height expansion nonuniform. Analyticity of the bulk integral in η^2 does not justify analyticity of the complete answer: terms linear in height or involving logarithms have not been excluded. A complete calculation at positive height would first determine whether the nonzero bulk response survives, cancels, or requires a different operator definition.

S9 Cumulative-storage gluing and joint continuation

This section gives the algebra behind the continuation criterion in the exposition [1]. It follows the inherited cumulative-storage note [19]. All operators here act on ordinary unweighted interval Hilbert spaces. The identities supply tests for a proposed construction; no arithmetic coupling bound or successful all-depth recursion is proved.

S9.1 The fixed-shift spatial Schur complement

After translating I_{L+h} to $(0, L+h)$, causality gives

$$V = V_{\omega, L+h} = \begin{pmatrix} X & 0 \\ Y & Z \end{pmatrix}, \quad X = V_{\omega, L}, \quad Z = V_{\omega, h}. \quad (\text{S9.1})$$

Here Y carries old input into the added output interval. Suppose $\|X\| < 1$ and $\|Z\| < 1$, and set

$$E = I - X^*X, \quad F_{\text{out}} = I - ZZ^*, \quad \mathcal{C} = F_{\text{out}}^{-1/2}Y E^{-1/2}.$$

These defects are coercive: each is bounded below by a positive multiple of the identity.

Proposition S9.1 (Cumulative gluing criterion). *Under these strict diagonal bounds,*

$$\|V\| \leq 1 \iff E - Y^*F_{\text{out}}^{-1}Y \succeq 0 \iff \|\mathcal{C}\| \leq 1. \quad (\text{S9.2})$$

Replacing both norm inequalities by strict inequalities gives the strict-contraction criterion.

Proof. The defect is

$$I - V^*V = \begin{pmatrix} E - Y^*Y & -Y^*Z \\ -Z^*Y & I - Z^*Z \end{pmatrix}.$$

Its lower diagonal block is coercive. Its Schur complement is

$$E - Y^*[I + Z(I - Z^*Z)^{-1}Z^*]Y = E - Y^*F_{\text{out}}^{-1}Y = E^{1/2}(I - \mathcal{C}^*\mathcal{C})E^{1/2}.$$

The middle identity follows by multiplying by $I - ZZ^*$. Schur elimination proves positivity and, using bounded invertible triangular elimination factors, equivalence of coercivity. $\square$

The output orientation $I - ZZ^*$ is essential; replacing it with $I - Z^*Z$ inside the displayed cross term changes the condition. At non-strict diagonal contraction the inverses need not exist, so this proof cannot be used without a range-compatible extension.

S9.2 Cayley coordinates and reflection

For $\omega > 0$ the finite-window transfer is Volterra convolution by a locally integrable kernel. It is quasinilpotent: conjugation by e^{-cx} gives a convolution norm bounded by $\int_0^L e^{-cu}|k_\omega(u)| du$, which tends to zero as $c \rightarrow \infty$. Each finite- c conjugation is a bounded similarity, so the spectral radius is zero. Hence $I + V$ has a bounded inverse without assuming contraction.

Define

$$\mathcal{Z} = (I - V)(I + V)^{-1}, \quad P_{\omega, L} = \frac{2}{\omega} \operatorname{Re} \mathcal{Z}, \quad \operatorname{Re} A = \frac{1}{2}(A + A^*). \quad (\text{S9.3})$$

Multiplying the real part on the two sides gives

$$I - V^*V = \frac{\omega}{2}(I + V^*)P_{\omega, L}(I + V). \quad (\text{S9.4})$$

Thus $P_{\omega, L} \succeq 0$ is exactly contraction; coercivity is exactly strict contraction in operator norm. For the real causal kernel, interval reflection $J_L f(x) = f(-x)$ on the centered interval gives $V^* = J_L V J_L$. Consequently $\mathcal{Z}^* = J_L \mathcal{Z} J_L$, and $P_{\omega, L}$ commutes with reflection. The raw input defect need not commute with it.

For the block split (S9.1), put $R_X = (I + X)^{-1}$, $R_Z = (I + Z)^{-1}$. Block inversion yields

$$P_{\omega, L+h} = \begin{pmatrix} A & B^* \\ B & F \end{pmatrix}, \quad A = P_{\omega, L}, \quad F = P_{\omega, h}, \quad B = -\frac{2}{\omega} R_Z Y R_X. \quad (\text{S9.5})$$

The two useful orientations of the diagonal identities are

$$A = \frac{2}{\omega} R_X^* E R_X, \quad F = \frac{2}{\omega} R_Z F_{\text{out}} R_Z^*. \quad (\text{S9.6})$$

Under the strict bounds of Proposition S9.1,

$$\|F^{-1/2} B A^{-1/2}\| = \|F_{\text{out}}^{-1/2} Y E^{-1/2}\|. \quad (\text{S9.7})$$

Indeed the left normalized block equals $-\mathcal{U}\mathcal{C}\mathcal{W}$, where

$$\mathcal{U} = \sqrt{2/\omega} F^{-1/2} R_Z F_{\text{out}}^{1/2}, \quad \mathcal{W} = \sqrt{2/\omega} E^{1/2} R_X A^{-1/2}.$$

The displayed diagonal identities make these invertible maps unitary. This proves (S9.7) without commuting any noncommuting factors. The Cayley change preserves the full coupling norm; it provides coordinates rather than stronger bounds.

S9.3 A step changing both length and shift

Fix the proposed append length h . At a reference positive shift α , suppose $A_\alpha = P_{\alpha, L}$ and $F_\alpha = P_{\alpha, h}$ are coercive. The enlarged block need not yet be positive. Let B_α be its cross block and put $\mathcal{K} = F_\alpha^{-1/2} B_\alpha A_\alpha^{-1/2}$. For a candidate new shift $\beta > 0$, define

$$H_A = A_\alpha^{-1/2} (A_\beta - A_\alpha) A_\alpha^{-1/2}, \quad (\text{S9.8})$$

$$H_F = F_\alpha^{-1/2} (F_\beta - F_\alpha) F_\alpha^{-1/2}, \quad (\text{S9.9})$$

$$H_B = F_\alpha^{-1/2} (B_\beta - B_\alpha) A_\alpha^{-1/2}. \quad (\text{S9.10})$$

Congruence by the reference diagonal metrics makes the enlarged form at β equal to

$$\begin{pmatrix} I + H_A & (\mathcal{K} + H_B)^* \\ \mathcal{K} + H_B & I + H_F \end{pmatrix}.$$

It is coercive if and only if $I + H_F$ is coercive and

$$\boxed{I + H_A - (\mathcal{K} + H_B)^* (I + H_F)^{-1} (\mathcal{K} + H_B) \succ 0.} \quad (\text{S9.11})$$

Here $\succ 0$ means a strictly positive operator lower bound. This is a second Schur complement; its signed changes retain improvements in either diagonal storage or cross coupling. A simple sufficient condition is

$$H_A \succeq aI, \quad H_F \succeq bI, \quad a, b > -1, \quad \|\mathcal{K} + H_B\|^2 < (1+a)(1+b).$$

Replacing each signed change by its absolute norm is more conservative and may discard the improvement that makes the step possible. All these statements concern operators, not only their finite compressions.

S9.4 Scope near zero shift and at increasing depth

The causal Cayley symbol has the core expansion

$$\frac{2}{\omega} \frac{1 - K_\omega(p)}{1 + K_\omega(p)} = a_0(p) + \omega^2 \left(\frac{a_0''(p)}{6} - \frac{a_0(p)^3}{12} \right) + O(\omega^4), \quad (\text{S9.12})$$

where primes mean p derivatives. To see the coefficient, write $\log K_\omega = -\omega a_0 - \omega^3 a_0''/6 + O(\omega^5)$ and expand the resulting hyperbolic tangent. The expression is even in ω ; its use after inversion is on appropriate smooth cores, not an operator-norm comparison with the unbounded central form. The extra cubic term also distinguishes accumulated storage from the instantaneous generator.

Positive-shift comparisons in (S9.11) avoid asserting uniform convergence on the central form domain. They still require rigorous estimates of the full blocks and their weak directions. An iteration must additionally prove $L_{j+1} = L_j + h_j$, $\sum_j h_j = \infty$ and $\omega_j \downarrow 0$. The algebra does not supply those estimates. Discarding output through a projection, assuming monotonicity in shift, or extrapolating local extension cannot replace them.

S10 Reproducible diagnostics and controls

The accompanying package contains three Python standard-library programs and their preserved JSON results. They register 268 finite cases. Each case records its name, numerical value, comparison direction, threshold and pass status. Replay requires identical parameters, case identities and thresholds and a passing result; small platform-dependent changes in floating values are permitted. No case uses zeta zeros or infers an arithmetic positivity theorem.

Program in numerics/	Cases	Main comparisons
<code>check_smooth_variation.py</code>	102	Inverse-map geometry, ordered ODE variation, endpoint motion, reparametrization, abelian and scalar-coupling controls.
<code>check_defect_endpoints.py</code>	99	Chiral projectors, all orientation signs, endpoint nullspaces, R charge and noncommuting adjoint transport.
<code>check_reflected_junction.py</code>	67	Reflected maps and color gluing, free Gram matrices, twist failures, smooth joins and independent bulk quadrature.

For transparency the Clifford matrices may be generated by five tensor factors. With $j = 0, \dots, 4$ and Pauli matrices σ_i ,

$$\Gamma_{2j} = \sigma_3^{\otimes j} \otimes \sigma_1 \otimes I^{\otimes(4-j)}, \quad \Gamma_{2j+1} = \sigma_3^{\otimes j} \otimes \sigma_2 \otimes I^{\otimes(4-j)}. \quad (\text{S10.1})$$

They satisfy the stated Clifford relations. Projector traces are compared with independent row reduction. Correct triplet assignments and the allowed orientation supply nonzero controls for the excluded choices. The adjoint check uses noncommuting Hermitian matrix segments, so a mistaken reversal of matrix order cannot pass merely because an abelian example commutes.

The transport checks compare direct variation of an ordered ODE solution with the integrated insertion formula on smooth matrix backgrounds, including mesh refinement. The reflected response checks compare the analytic coefficient with a separately integrated double integral and check

finite Gaussian current Gram matrices. The flat and ordinary-adjoint cases are positive or zero controls for the claimed failures. These comparisons support the written derivations; they do not certify physical supersymmetry transformations, quantum operator existence, continuum reflection positivity, all-order Ward identities or a stochastic contour limit.

The standalone manuscript package can be checked with

```
python3 validation/drafts.py check -replay
```

The live investigation additionally has `validation/check_package.py`, which inventories the research notes and their inherited repository sources. That broader inventory is separate from the portable manuscript snapshot.

The auxiliary free-kernel background in Sections S1 and S2 is derived explicitly in the text. The 268 count refers to this investigation’s three programs, not to every calculation in its parent investigations. Hash and build records establish file identity and reproducibility, not mathematical correctness.

S11 Provenance and contribution disclosure

This companion was separated from the unified Wilson–Loewner manuscript on 20 September 2026 by OpenAI GPT-6 (Codex), for Edward Baker. Version 0.3 shares the exposition’s version number to identify the matched pair. The configured reasoning effort was not exposed to the assistant in this session and is not inferred. Both documents are working research drafts without independent mathematical review.

Sections S1–S8 preserve the detailed arithmetic, free-field, contour, supersymmetry, endpoint and reflection calculations from versions 0.1–0.2. The three principal local source notes are *Smooth Wilson variation and the constant Loewner driver*, *Defect-compatible contour freedom and the endpoint pairing*, and *Physical reflection, the reference junction, and a first bulk response*, all dated 19 September 2026. The parent Wilson-lines work supplies the angular regulator and auxiliary Gaussian model; the Loewner and fractional-dimension investigations supply the transfer factorization and scoped realization restrictions. Section S9 transcribes and explains the inherited cumulative continuation algebra [19]; it supplies no new coupling bound. The full program and hypotheses are in the exposition [1].

The original assembly checked the printed conventions for Loewner normalization, tangent and D3–D5 projectors, Baker endpoint transformations and semicircle, Dorn–Pershin reflection, and the Weil-form weights against the cited primary papers. Version 0.3 does not claim a new exhaustive literature or priority survey. No novelty is claimed for standard transport variation, tangent-coupled supersymmetry, the ray endpoint selection rule, or the classical Weil criterion.

`MANUSCRIPT_SOURCES.json` records assembly and revision inputs. `BUILD_RECORD.json` binds both documents and supporting checks; each dated archive has a `SNAPSHOT.json`. These are identity records, not proof certificates. The historical unified drafts remain unchanged. Every TeX input needed to build either current document is in this package; no shared-background or parent-investigation TeX file is loaded at build time.

References

- [1] Drafted for E. Baker by OpenAI GPT-6 (Codex), *Shifted-zeta evolution and Wilson lines: a Loewner realization program for cumulative positivity*, version 0.3, 20 September 2026. Main working exposition, `manuscript.pdf` and `manuscript.tex`, in the same package.

- [2] J. M. Maldacena, *Wilson loops in large N field theories*, Physical Review Letters **80** (1998), 4859–4862, [arXiv:hep-th/9803002](#).
- [3] K. Zarembo, *Supersymmetric Wilson loops*, Nuclear Physics B **643** (2002), 157–171, [arXiv:hep-th/0205160](#).
- [4] O. DeWolfe, D. Z. Freedman and H. Ooguri, *Holography and defect conformal field theories*, Physical Review D **66** (2002), 025009, [arXiv:hep-th/0111135](#).
- [5] D. Gaiotto and E. Witten, *Supersymmetric boundary conditions in $\mathcal{N} = 4$ super Yang–Mills theory*, Journal of Statistical Physics **135** (2009), 789–855, [arXiv:0804.2902](#).
- [6] E. B. Baker III, *Supersymmetric open Wilson lines*, Journal of High Energy Physics **03** (2011), 140, [arXiv:1102.4948](#).
- [7] G. F. Lawler, *Schramm–Loewner evolution (SLE)*, lecture notes, especially Lecture 2, Proposition 2.17 and equation (2.8), [author’s PDF](#), accessed 19 September 2026.
- [8] M. Bauer and D. Bernard, *Conformal field theories of stochastic Loewner evolutions*, Communications in Mathematical Physics **239** (2003), 493–521, [arXiv:hep-th/0210015](#).
- [9] K. Osterwalder and R. Schrader, *Axioms for Euclidean Green’s functions*, Communications in Mathematical Physics **31** (1973), 83–112, [doi:10.1007/BF01645738](#).
- [10] H. Dorn and V. D. Pershin, *Concavity of the $Q\bar{Q}$ potential in $\mathcal{N} = 4$ super Yang–Mills gauge theory and AdS/CFT duality*, Physics Letters B **461** (1999), 338–344, [arXiv:hep-th/9906073v3](#).
- [11] M. Suzuki, *Weil’s quadratic form via the screw function*, preprint (2026), version 2, [arXiv:2606.09096v2](#). Used for the classical compact-support Weil criterion and conventions; its proposed limiting spectral construction is not assumed here.
- [12] Riemann Hypothesis research program, *Deformation flows for the shifted Weil family: the transfer as an all-pass filter*, working draft 0.8, 17 September 2026, and endpoint continuation notes, 18–19 September 2026. Repository investigation [papers/susy-positivity/investigations/wilson-lines/](#). Internal research, with model-assisted contributions disclosed in the source documents.
- [13] OpenAI GPT-6 (Codex), for Edward Baker, *Angular smearing, the finite difference form, and a Robin model of the shift*, research note, 19 September 2026, in the Wilson-lines investigation, [notes/ANGULAR_SMEARING_AND_ROBIN_MODEL_20260919.md](#).
- [14] Riemann Hypothesis research program, *The Markov part of the shifted Weil transfer*, working draft 0.3, 18 September 2026. Repository investigation [papers/susy-positivity/investigations/loewner/](#). Internal research; its direct realization proposal was closed within the tested scope.
- [15] Riemann Hypothesis research program, *A scattering matrix without a space*, working draft 0.1, 18 September 2026. Repository investigation [papers/susy-positivity/investigations/fractional-dimension/](#). Internal research; the continued lattice count and its limitations are discussed in its Section 6.

- [16] Riemann Hypothesis research program, *Shared background for the supersymmetric positivity program*, repository source `papers/susy-positivity/background_section.tex`, state consulted 19 September 2026. In particular, the boxed transfer and defect equations beside the energy derivative and the sufficient depth-shift implication.
- [17] OpenAI GPT-6 (Codex), for Edward Baker, *Prescribed shift evolution, cumulative storage, and a depth-shift continuation*, research note, 19 September 2026, `notes/SHIFT_FLOW_CUMULATIVE_STORAGE_AND_CRITICAL_PATH_20260919.md` in the Wilson–Loewner investigation.
- [18] Riemann Hypothesis research program, *Depth extension with an adaptive shift*, working research, 10 September 2026, `papers/shifted-zeta/storage-depth/archive/background/1-critical_path_research.md`. Internal, model-assisted research; independent review remains open.
- [19] Riemann Hypothesis research program, *A depth-shift path controlled by cumulative storage*, working research, 10 September 2026, `papers/shifted-zeta/storage-depth/archive/background/2-cumulative_storage_path.md`. Internal, model-assisted research; the continuation bounds are conditional.
- [20] E. Baker, *Residual-controlled depth extension of finite-horizon Weil positivity*, working draft 0.3, with model assistance disclosed there; see Appendix A, `papers/shifted-zeta/storage-depth/manuscript/storage_depth.tex`. The recorded enclosures use that paper’s working normalization; independent normalization verification and review remain open.
- [21] M. Suzuki, *A canonical system of differential equations arising from the Riemann zeta-function*, RIMS Kokyuroku Bessatsu **B34** (2012), 397–435, especially Section 1.4, [publisher’s PDF](#).